Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

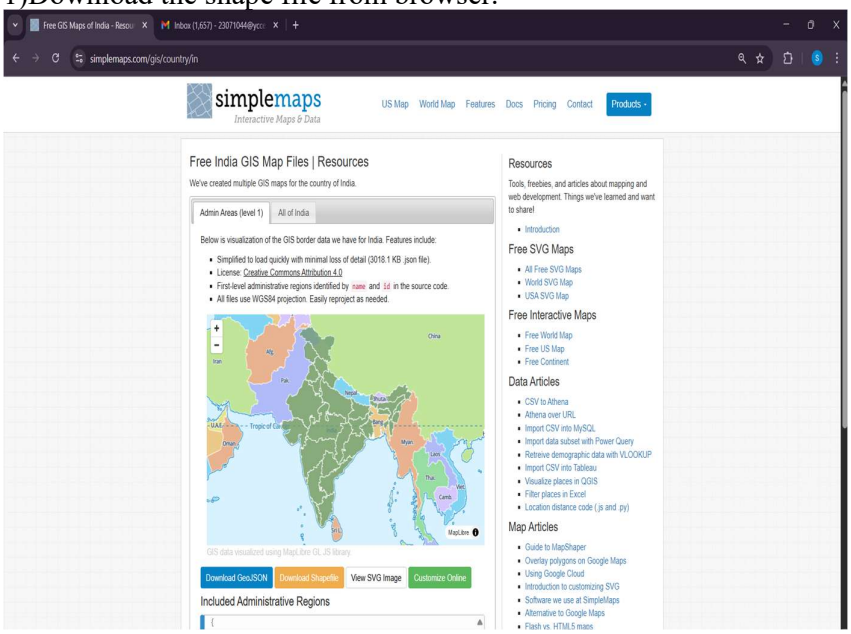
Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**02**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Download a shapefile, inspect metadata, and reproject it to a different CRS.
Problem Definition	The task involves downloading a shapefile, inspecting its metadata to analyze spatial reference and attribute information, and accurately reprojecting it to a different Coordinate Reference System (CRS).
Theory (100 words)	In Geographic Information Systems (GIS), a **shapefile** is one of the most common formats used to store spatial information, including vector features such as points, lines, and polygons. Rather than being a single file, a shapefile is composed of multiple linked files that together represent both the spatial geometry and related attribute information. To make the dataset more understandable and usable, **metadata** is included, which describes details such as the coordinate reference system (CRS), spatial coverage (extent), and attribute structure. The CRS plays a crucial role because it specifies how the digital map coordinates correspond to actual positions on the Earth's surface. When datasets use different CRSs, a process called **reprojection** is required to convert them into a common reference system. This step is essential for overlaying, integrating, and analyzing spatial data from various sources accurately.
Procedure and Execution (100 Words)	Implementation Steps: 1. Download the shapefile of India using a browser. 2. The dataset will be in a ZIP file; extract it to a folder. 3. Open **QGIS** on your computer. 4. In QGIS, go to **Layer → Add Layer → Add Vector Layer**, then browse to the extracted folder. 5. Choose the file with the .shp extension and click on 'Add' to load it. 6. The map of India with state boundaries will appear on the screen. 7. Right-click on the India shapefile layer, select 'Properties', then go to the 'Source' tab to view its current Coordinate Reference System (CRS). 8. To permanently reproject, right-click the layer → **Export → Save Features As…**.

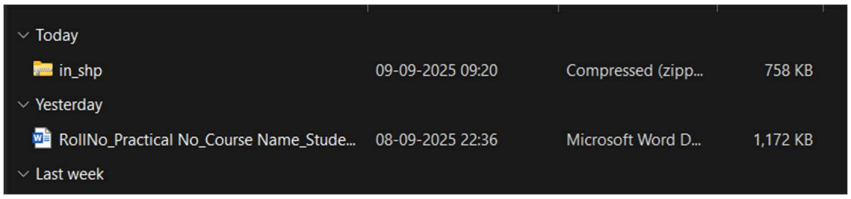
9. In the export window, choose the desired **CRS** (for example, **EPSG:4326 – WGS 84**).
10. Save the new shapefile and load it into QGIS to confirm that the CRS has been successfully changed.

Stepwise Screenshots with steps:

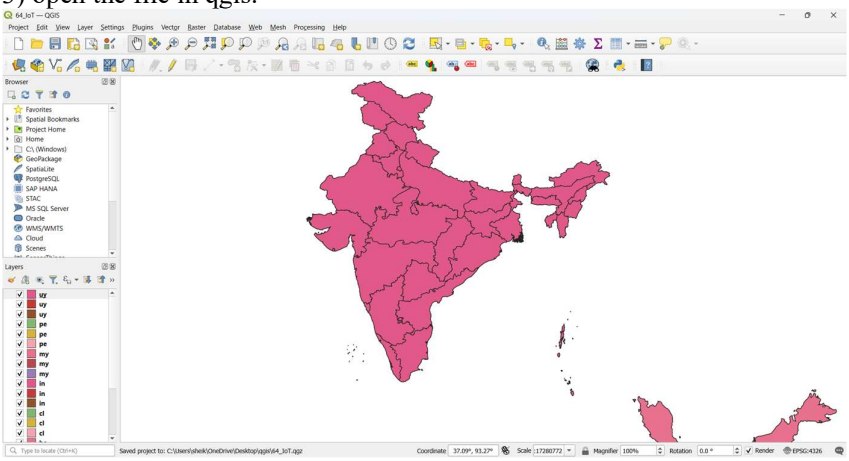
1)Download the shape file from browser.



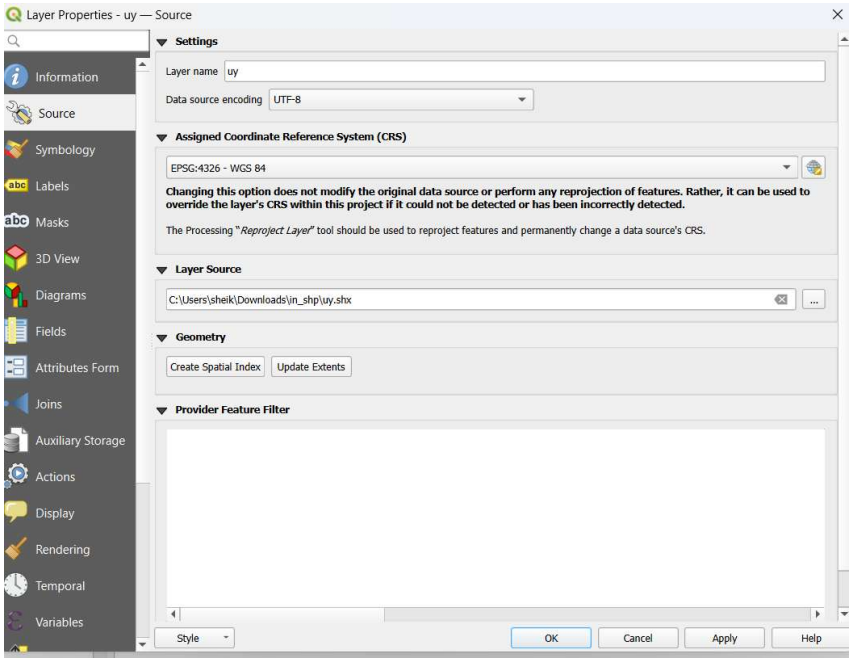
2) Extract the zip file

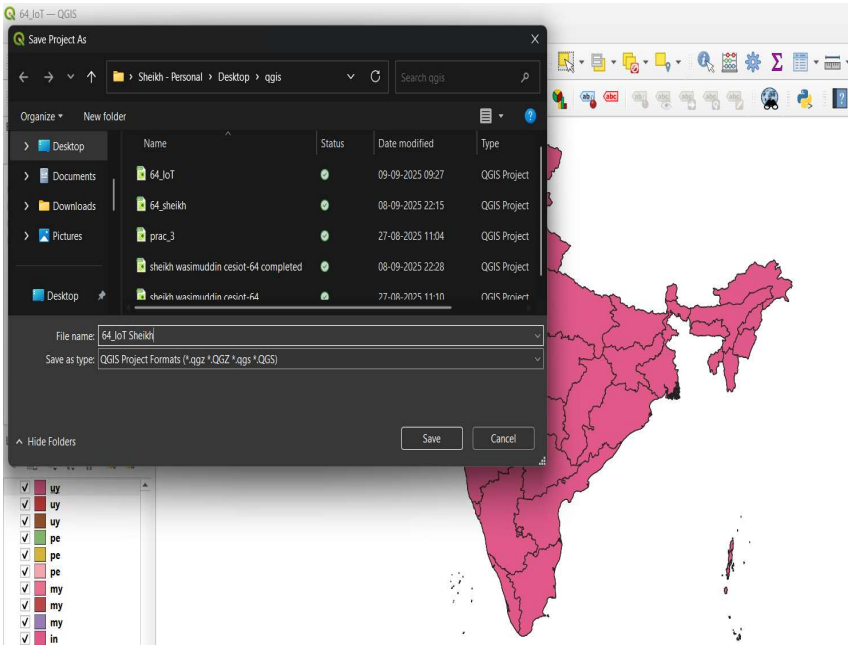


3) open the file in qgis.



4) Right-click on the India shapefile layer, select 'Properties', then go to the 'Source' tab to view its current Coordinate Reference System (CRS).



	5) To permanently reproject, right-click the layer → **Export** → **Save Features As….** 
Output Analysis	After completing the procedure, the shapefile of India was successfully visualized in QGIS with clearly defined state boundaries. Both the attribute information and metadata were reviewed, verifying the dataset's original CRS. Following reprojection into a new CRS, the map was rendered correctly without any distortions, indicating that the transformation was accurate. The layer properties confirmed the updated CRS, and all state features along with their attribute details remained intact. This ensures that the dataset is now properly prepared for precise spatial analysis and can be seamlessly integrated with other geospatial layers.
Conclusion	The practical task was carried out successfully. The shapefile of India was obtained, extracted, and loaded into QGIS. By reviewing the metadata, the spatial reference system and attribute details of the dataset were understood. The dataset was then reprojected to a different CRS, and the resulting map verified that the new projection was applied accurately without any data loss. This highlights the significance of proper CRS management in GIS to ensure precision and interoperability during spatial analysis.

Plag Report (Similarity index < 12%)	
Date	29-7-2025